Experiment No. 2

Name of experiment:

A. *Open circuit characteristics and magnetization characteristic of* ***separately excited D.C generator***.

B. *Open circuit characteristics and magnetization characteristic of* ***self-excited D.C generator***

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1. Object:

This Test is done to obtain magnetization characteristic and no load curve of separately exited D.C generator.

2. Theory:

The induce voltage in D.C machine is given by equation:

$$E_g = \frac{PZ\,\Phi N}{60A} \quad \ldots\ldots\ldots\ldots\ldots\ldots (1)$$

Where

N = speed in (r.p.m)
Z = number of conductors
Φ = air gap flux per pole in weber
P = number of pole
A = number of parallel paths

- *As* Z, P & A *are constants for certain machine, and then equation (1) can be written as*

$$E_g = C\Phi N \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (2)$$

- *Where C = constant*

No-Load Characteristics

It is represented by the relationship between the no-load terminal voltage and the field current .It is also called the magnetization curve or the open circuit characteristics. This can be obtain at constant speed .The shape of the curve is similar to B-H curve of the magnetic material of the magnetic circuit

The shunt generator may be excited either:

i. By connecting the field winding to a separated supply, thus is called separately excited generator.
ii. By connecting the field winding and the armature winding in parallel .It is called self-excited D.C shunt generator.

3. Procedure:

1. Connected the circuit as shown in fig (1).
2. Apply the rated voltage of the electrical motor prime mover and start it till it reaches to reaches to the rated speed.
3. Record the voltage induced due to the residual flux.
4. Change the voltage of the resistance to change the field current $\mathbf{I}_f$ step by step Up and Down.
5. Record the voltage induced on the table (1). And n=1400r.p.m.

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A. *Open circuit characteristics and magnetization characteristic of **separately excited D.C generator***.

Table No. (1) No load characteristic

I_f (A)	0	0.1	0.2	0.3	0.4	0.5	0.6
E_o (V)							

Table No. (2) Magnetization characteristic

If (A)	0	0.1	0.2	0.3	0.4	0.5	0.6	0.5	0.4	0.3	0.2	0.1	0
Eo (V)													

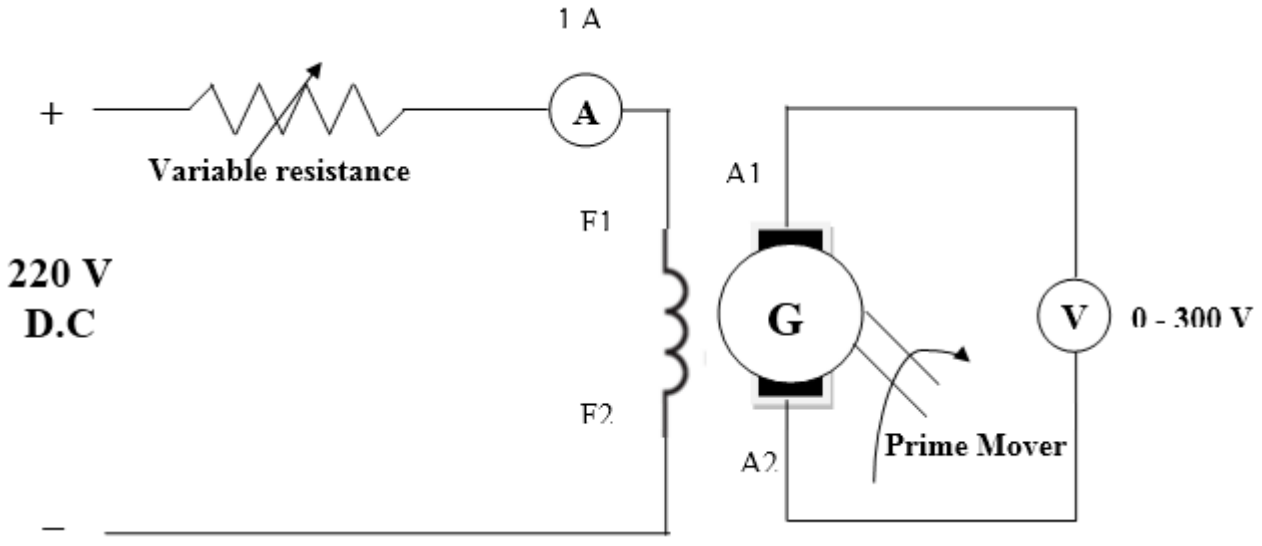


Figure (1)

B. *Open circuit characteristics and magnetization characteristic of* ***self-excited D.C generator***

Table No. (1) No load characteristic

I_f (A)	0	0.1	0.2	0.3	0.4	0.5	0.6
E_o (V)							

Table No. (2) Magnetization characteristic

If (A)	0	0.1	0.2	0.3	0.4	0.5	0.6	0.5	0.4	0.3	0.2	0.1	0
Eo (V)													

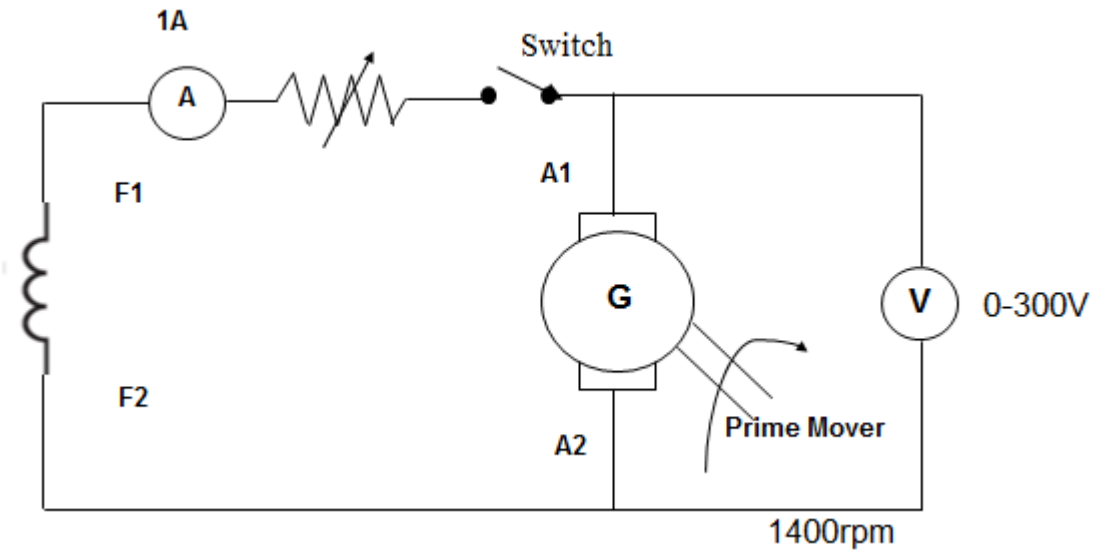


Figure (1)

Report Part A and B:

1. Plot the relationship between E_o & I_f (no-load char & magnetization characteristic). For separately and self-excitation.
2. How can the magnetization be neglected from the polarities?
3. What is the purpose of plotting the magnetization curve?
4. What different between separately exited and self-exited in this experiment A and B?
5. Discuss your results clearly and briefly.

